Xu Han

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Photonics M.Sc. student and LMU astrophysics graduate pursuing a career in astronomical instrumentation. My coursework and labs cover physical optics, lasers and fibre optics. My independent projects include Shack–Hartmann AO simulation and reconstructing ESO's OWL telescope design in Zemax.

Education

M.Sc. Photonics *Mar 2026 - Present*
Hochschule München (Munich University of Applied Sciences)
Remaining requirements are the project study and master's thesis. Expected completion in 2027. **Selected coursework:** Physical Optics and Modelling (**Zemax optical design**); Micro and Fiber Optics; Optical Measurement Technology; Laser Physics — Non-linear Optics — Ultrafast Optics; Semiconductor Optics and Components.

M.Sc. Astrophysics | LMU Munich *Apr 2022 - Aug 2024*
Grade: 1.55 (German scale). **Selected coursework:** Essentials of Advanced Astrophysics (1.3); Cosmology and Structure Formation (1.0); Statistical Methods; Instrumental Laboratory Courses. Degree certificate

B.Sc. Physics | University of Duisburg-Essen *Oct 2018 - Mar 2022*
Grade: 1.8 (German scale). Degree certificate

B.Eng. Electrical Engineering and Automation *Sep 2013 - Jul 2017*
Hubei University of Technology
Electronics, control systems and embedded systems.

Selected Research and Engineering Projects

Shack-Hartmann Wavefront Sensing & Adaptive Optics Simulation

Independent project | Interactive explorer | Source code *Mar 2026 - Present*

- Developed a modular Python SCAO framework connecting atmospheric turbulence, detector-level Shack-Hartmann sensing, calibrated reconstruction, deformable-mirror correction and delayed feedback.
- Traced four faint spots in a simulated 42 m telescope image to waffle modes the sensor barely detects. Filtering those modes reduced their mean brightness from 10.8 to 0.91 times the local halo level.
- Diagnosed a reconstruction failure associated with measurement dropout in a 2.2 m NGS model; stronger TSVD regularisation kept reconstruction usable throughout the matched two-second run.
- Cross-checked selected optical components against HCIPy and analytic references; documented reproducible experiments, model limitations and an interactive explorer of recorded results.

Optical Reconstruction of ESO's 100 m OWL Telescope *2026*

Independent extension of a Zemax laboratory project

- Rebuilt the proposed OverWhelmingly Large Telescope from ESO's Blue Book prescription. At 5 arcmin off axis, the RMS wavefront error was 1.472 µm, within 0.3% of ESO's 1.476 µm.
- At 3 arcmin, traced a 27-percentage-point discrepancy in ensquared energy to the focal surface. A spherical surface brought the model within 0.4 percentage points of ESO's sampled curve.

Weak Lensing Void Finder *Mar 2023 - Aug 2024*

Master's thesis | LMU Munich Supervisors: Dr. Chris Davies and Prof. Joseph Mohr.

- Developed a watershed void finder for 2D full-sky weak-lensing maps and tested its parameter sensitivity. Measured void abundance, effective radii, ellipticities and convergence profiles in DES Y3 data.
- Trained Gaussian-process emulators (scikit-learn) on simulations of 2,500 cosmologies and used them with MCMC for parameter inference.

Experimental Collisional Charging Apparatus *Mar 2021 - Aug 2021*

Bachelor's thesis | University of Duisburg-Essen Supervisors: Prof. Gerhard Wurm and Dr. Felix Jungmann.

- Designed the mechanical apparatus in CAD, fabricated components by 3D printing and assembled the setup; integrated and debugged electronic charge-measurement modules.
- Conducted measurements and analysed grain-size dependence of collisional charging, testing a dust-collision growth hypothesis relevant to early planet formation.

Laboratory and Instrumentation Training

- **Astronomical instrumentation laboratory:** Closed the Shack–Hartmann loop on LMU's ALPAO bench in July 2022, first correcting static aberrations with the deformable mirror, then correcting turbulence introduced by a screen. Other labs covered CCD imaging and reduction, optical interferometry and radio measurements of Galactic rotation.
- **Photonics laboratory:** Measured laser beam quality (M^2), shaped ultrashort pulses and characterised them by autocorrelation. Fibre experiments covered OTDR, polarisation, Bragg gratings and coupling. Also worked on APD measurements, laser spectra, laser interferometry and nonlinear optics.
- **Observatory experience:** Took and processed images with my supervisor at Wendelstein (43 cm).

Additional Independent Projects

Astrophotonics

- Modelled how AO wavefront error, injection jitter and photonic-lantern mode count affect fibre coupling, spectral SNR and radial-velocity precision.

Work Experience and Career Break

Python Developer | BaryonBib *Dec 2024 - Jan 2025*

MAX!mize start-up team, MPA, Garching

- Implemented and tested LLM features for the team's centralised Python bibliography platform.

Medical leave *Jan 2025 - Dec 2025*

Recovery and rehabilitation following an accident and surgery.

Talks, Workshops and Schools

MAVIS 2026 workshop | Participant *Sep 2026*

Tonale Winter School on Cosmology | Participant *Dec 2024*

ORIGINS Lensing Day *Nov 2024*

Contributed talk: Watershed on the curved sky

Cambridge-LMU Cosmology Meeting | Participant *Dec 2023*

Advanced Astrophysical Seminar | LMU Munich *Nov 2022*

Exploring Telescope Optics: How Do the Largest Optical Systems Work? A 45-minute presentation followed by 15 minutes of questions. Supervisor: Dr. Frank Grupp.

Technical Skills

Optical modelling: Wavefront sensing, Fourier PSFs, calibration, regularised reconstruction and closed-loop diagnostics; HCIPy and TIPTOP.

Zemax OpticStudio: telescope prescription reconstruction, PSF and ensquared-energy analysis

Scientific computing: Python (NumPy, SciPy, Matplotlib, pandas, scikit-learn); MATLAB; Mathematica; statistical analysis and MCMC.

Experimental methods: CAD, 3D printing, mechanical assembly, electronic measurement integration, optical alignment and interferometry.

Research tools: Linux, Git, Jupyter and LaTeX.

Languages and Interests

Languages: Chinese (native); English (fluent); German (fluent); French (beginner).
Interests: Astronomical photography and alpine mountaineering.

Academic References

Dr. Chris Davies — direct supervisor for my M.Sc. thesis.
Prof. Joseph Mohr — M.Sc. thesis supervisor. Taught Cosmology and Structure Formation and co-taught Essentials of Advanced Astrophysics.